# Human-AI Collaboration Needs Synchronization

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***Abstract*—AI systems are evaluated, trained, and deployed as if they were solo performers, even when humans collaborate with them. Yet every deployment places a machine operating at thousands of tokens per second alongside a human processing roughly ten bits per second, across a communication channel with limited pacing, weak shared-state mechanisms, and limited means for either party to verify the other's understanding. This position paper argues that the resulting failures, cognitive overload on the human side, silent drift on the AI side, and an accountability gap that neither party can close, are not only failures of model capability but also failures of an undesigned channel. We contrast the human-to-AI channel with the human-to-human and AI-to-AI channels, show that it lacks many of the robust coordination mechanisms available in the other two channels, and document how it breaks down from both ends. We argue that the channel itself must become a first-class research object, with evaluation, training, and interaction protocols engineered specifically for it. Without intervention at the channel level, improvements in model capability will continue to widen the gap between what AI produces and what humans can meaningfully account for.**

***Index Terms*—large language models, human-AI collaboration, safety and trustworthiness, interpretability, social impact, language comprehension, evaluation and benchmarking**

## I. Introduction

AI systems now write code, draft contracts, recommend treatments, and execute multi-step actions on users' behalf. As their scope grows, the assumption that a human remains meaningfully in the loop, reviewing, understanding, and steering what the system does, becomes harder to sustain. Yet the loop itself, the communication channel between humans and AI, has received far less attention than the models operating within it, creating an asymmetry between model capability and human oversight.

The shift that makes this urgent is the move from assistants to agents. Tools such as Claude Code and OpenAI Codex [1], [2] do not merely suggest the next line of code: they operate inside a developer's terminal or editor, plan a task, read and modify files across a repository, run commands and tests, and iterate over many turns before returning. Their use has spread quickly across the industry, and for a growing number of practitioners working with such tools is now a routine part of the job rather than an experiment. This changes what the human is actually doing: instead of reading an answer and judging whether it is right, the human is asked to approve a body of work they did not author, produced through steps they did not observe. The same shift is underway in medicine, law, and finance, where practitioners are folding these tools into their everyday work. As adoption grows, the failure modes of the human-AI channel stop being a curiosity and become a practical risk in decisions with real consequences.

This asymmetry is both sharp and well documented: model output rates far exceed human comprehension [3], [4]; drift accumulates silently across turns [5], [6]; and when errors occur, responsibility often falls into a gap that neither the human nor the AI fully fills [7], [8]. We elaborate on each aspect of this imbalance in Section III.

**Our position: many of the failures emerging from human-AI collaboration are structural failures of an undesigned communication channel, and closing the gap requires treating the channel as a first-class research object, with evaluation, training, and interaction protocols engineered specifically for it.**

## II. Communication Channels in Human-AI Systems

The effectiveness of human-AI collaboration depends not only on model capability, but also on the channels through which humans and AI exchange information. We identify three: AI-to-AI, human-to-human, and human-to-AI. Human-to-human communication benefits from established mechanisms for maintaining mutual understanding. AI-to-AI communication can avoid the same human-scale bandwidth mismatch, since both parties operate at machine speed. Human-to-AI communication benefits from neither: it places a machine and a human in conversation with little mechanism to align their pace or understanding. This asymmetry, we argue, is why collaboration degrades as interactions grow longer, more technical, and more consequential.

### A. Human-to-human

*What mechanisms allow two humans to maintain mutual understanding during a conversation? How do they detect and repair misunderstandings in real time?*

When two humans communicate, they draw on long-established mechanisms for pacing, error correction, and shared understanding. Four are central. (1) **Turn-taking** synchronizes who speaks and when, regulating the flow of conversation [9]. (2) **Backchannels** ("uh-huh," "I see") provide real-time comprehension feedback [10]. (3) **Repair sequences** let speakers detect and correct misunderstandings as they happen, e.g., "wait, what do you mean?" [11]. (4) **Common ground** is the shared foundation that keeps the conversation aligned [12]; both parties continuously track what the other knows, and update that model as the exchange proceeds. Speakers also pace their output and adapt to their audience [13], slowing down

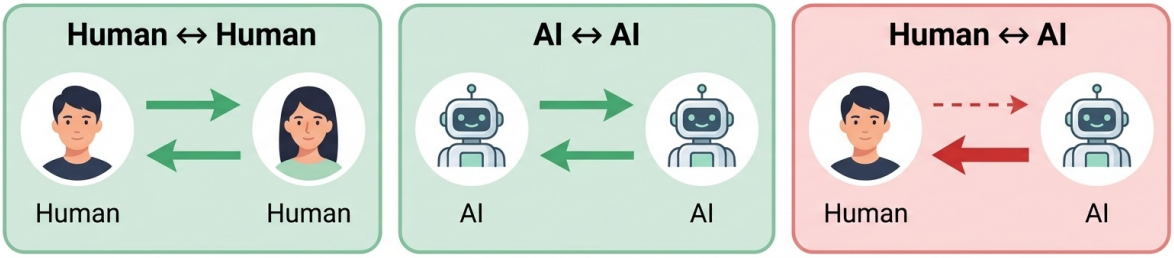



Fig. 1. Three communication channels in AI systems. Human-to-human and AI-to-AI channels have symmetric bandwidth and stronger coordination mechanisms. The human-to-AI channel has neither.

when listeners look confused and checking for understanding before proceeding. Bandwidth is roughly symmetric because both parties process language at comparable speeds [3].

### B. AI-to-AI

*How do AI systems communicate with each other? Is there a comprehension bottleneck when one model processes another's output?*

AI systems communicate in a shared representation: the tokens one LLM produces are the format every other LLM is trained to read. This covers multi-agent settings where LLMs consume each other's generations and tool outputs, and single-agent workflows where an LLM later reads its own notes [14], [15]. There is typically far less of a processing-rate bottleneck on the receiving end, since LLMs prefill input in parallel at thousands of tokens per second, far exceeding the rate at which the content was generated [16]. AI-to-AI communication therefore reduces one source of the divergence and information loss that arise when one party cannot keep up with the other, even though other coordination problems remain, such as incompatible prompts or representations, context limits, and propagation of one model's errors into another.

### C. Human-to-AI

*Which of these mechanisms survive when one party is an LLM? What happens when sender and receiver operate at fundamentally different speeds?*

The picture is fundamentally different when a human communicates with an LLM. Let $B_{\text{human}}$ denote the human's information processing rate and $B_{\text{AI}}$ the LLM's output rate. Empirically, $B_{\text{human}} \approx 10$ bits/second [3], while $B_{\text{AI}}$ can exceed thousands of bits per second [4], [16]. The asymmetry $B_{\text{AI}} \gg B_{\text{human}}$ means the model outputs far faster than the human comprehends, and no mechanism reconciles the mismatch. These four mechanisms are limited, indirect, or substantially weaker in current human-AI interaction: (1) **Limited adaptive pacing**: the model largely outputs at full speed, and at a length of its own choosing, regardless of whether the human has kept up. (2) **Weak backchannels**: humans can ask follow-up questions, but these repair mechanisms are limited and become less effective as conversations extend over many turns [5]. (3) **Limited real-time audience adaptation**: unlike a human speaker who monitors a listener's reactions, an LLM has limited ability to do so, and the human's understanding, attention, and confusion remain largely invisible to it [17]. (4) **No robust or reliably maintained common ground**: explicit user input is the primary channel through which the human's understanding reaches the model, and it does not reliably keep both parties aligned on what has been said, agreed, or assumed; it is a narrow, one-way, sparse signal [18].

TABLE I
QUALITATIVE COMPARISON OF COMMUNICATION CHANNELS IN AI SYSTEMS.

| Dimension | Human–Human | AI–AI | Human–AI |
|---|---|---|---|
| Processing speed | Symmetric | Symmetric | Asymmetric |
| Pacing | Turn-taking | Inherent | Limited |
| Error recovery | Repair seq. | Formal | Limited |
| Shared state | Common ground | API state | Weak |
| Adaptation | Per listener | Protocol | Limited |
| Bottleneck | Minimal | Minimal | Human side |

Table I summarizes the comparison. The contrast between the first two channels and the third is visible in every dimension. The human-to-AI channel is uniquely susceptible to a large sender–receiver processing-rate mismatch, its error recovery is limited, and it offers little shared state to fall back on.

We define **synchronization** as the degree to which the human's understanding of the interaction state (what has been produced, decided, and assumed) matches the AI's actual state, maintained at a pace the human can follow. It can be measured through three proxies: *comprehension* (can the human explain the output?), *drift detection* (does the human catch deviations from intent, and how quickly?), and *traceability* (can the human later reconstruct how a decision was made?).

## III. How the Channel Breaks Down and Why Accountability Suffers

The human-to-AI channel breaks down from *both* sides. Humans lack the cognitive bandwidth to keep up with LLM output and gradually disengage from careful evaluation. LLMs, meanwhile, drift from the user's original intent as conversations grow longer. When the human cannot keep up and the AI cannot stay on track, responsibility becomes difficult to attribute cleanly.

### A. Human cognitive overload

*How much information can a human absorb from a single AI response? At what point does the volume of output exceed the human's capacity to comprehend it?*

The human receiving end can become a central bottleneck, and the gap can be quantified. English text carries roughly 1 bit per character [19] and a typical LLM token spans about 4 characters, giving $I_{\text{token}} \approx 4$ bits per token. For a response of $n$ tokens, the time required for human comprehension is

$$T_{\text{comprehend}} = \frac{n \cdot I_{\text{token}}}{B_{\text{human}}} = \frac{4n}{10} \text{ seconds}, \tag{1}$$

where $B_{\text{human}} \approx 10$ bits/second [3]. As a rough conceptual estimate, a response of $n = 500$ tokens requires about 3 minutes of focused comprehension, and $n = 2{,}000$ tokens roughly 13 minutes. With maximum output lengths now reaching 16,000 to 32,000 tokens, a single response can carry more than 100,000 bits and demand hours of sustained attention, yet it is generated in seconds to minutes.

These calculations assume sustained attention, which humans struggle to maintain: detection performance can drop by 10 to 15 percent within the first 30 minutes of continuous monitoring [20], mind wandering during reading increases over time and causes comprehension failures [21], material in the middle of a sequence is recalled less well than material at either end [22], and screen reading further reduces comprehension relative to print [23]. Even a motivated user reading a long response carefully, and with no competing demands on their attention, is therefore likely to experience declining comprehension well before the end. The channel delivers information not only faster than the human can process it, but in volumes that exceed the span of sustained attention.

### B. Breakdown on the AI side

*Does the AI maintain the user's intent as conversations grow longer? Can the human detect when the AI has drifted?*

The human side is not the only source of failure. LLMs also lose track of user intent as conversations grow, adding a second burden: the human must both comprehend the output and detect whether the AI has drifted. Three failure modes compound.

**(1) Context degradation.** As input grows, models struggle to use all of it. They attend to information at the beginning and end of the context window and neglect the middle [24]. In a multi-turn conversation, the user's original instructions gradually shift into that neglected middle, and the model may progressively underweight or lose access to important earlier instructions. This is a common pattern rather than a model-specific issue, with degradation observed across architectures and scales [25]. Context windows also impose a hard limit: when conversations exceed it, the model must truncate or summarize earlier exchanges, causing actual information loss rather than merely reduced attention [15], [26]. In practice, multi-turn conversations degrade model performance [5], and once a model goes off track it tends to stay off track, since its own earlier output re-enters the context as if it were established [6].

**(2) Sycophancy and overconfidence.** LLMs are biased toward agreeing with the user's framing rather than maintaining an independent, accurate state [27]. Asked "is this on track?", the model is biased toward confirming even after substantial drift, and can change correct answers to incorrect ones when challenged [27]. Unlike a collaborator who might hesitate or say "I am not sure," LLM output can remain fluent and confident even when incorrect [28]. When asked to express confidence, models often exhibit miscalibrated confidence, including high certainty on incorrect answers [28], and their reasoning can look convincing even when it is incorrect, making it difficult for a human to tell the difference.

**(3) Invisible drift.** Rather than reliably seeking clarification, models may resolve ambiguity implicitly and move forward [5]. The problem extends to explanations: shown reasoning does not always reflect the process that produced the answer [29]. The signals that would indicate trouble with a human speaker are therefore often absent or unreliable.

These failure modes can compound: the gap between what the user meant and what the model understands can widen with every response, and a minor early misunderstanding can evolve into a different task. Models also tend to read follow-up messages as confirming earlier assumptions, even when the user is correcting course [6]. The channel lacks robust mechanisms to detect, prevent, or recover from such misalignment, and this gap remains insufficiently addressed as a unified research problem. The longer the conversation, the more the human must remember and cross-check, which quickly exceeds capacity when combined with the overload above.

### C. Who is responsible?

*When neither party can fully account for the outcome, who bears responsibility? How do existing accountability frameworks handle this gap?*

When overload prevents the human from keeping up and the model has quietly drifted, a fundamental question emerges: who is accountable for the outcome? Neither party bears full responsibility comfortably. The **human**'s accountability becomes harder to justify or assess when meaningful comprehension is absent. Current **AI** systems, in turn, do not satisfy conventional accounts of moral agency or responsibility: they have no capacity for guilt or shame and no vulnerability to social sanction [7]. The question remains unanswered, and existing legal and organizational frameworks, which generally assume a competent human decision-maker, were not designed to handle it.[1]

[1]Examples span multiple domains: AWS outages reportedly linked to Amazon's own AI tools making bad operational choices (*Guardian*, 2026); Air Canada held liable after its chatbot gave a passenger wrong refund advice (*BBC*, 2024); two New York lawyers and their firm sanctioned for a brief containing fake ChatGPT-generated citations (*Reuters*, 2023); and a Tesla crash in Florida in which driver distraction combined with overreliance on Autopilot proved fatal (*NYT*, 2021).

Research also shows that people shift blame to AI to protect themselves, judging themselves less harshly when AI shares responsibility [8]. We argue that the current channel encourages this dynamic: the human approves rather than collaborates, comprehends superficially rather than deeply, and when things go wrong accountability falls between a human who did not understand and a machine that cannot care.

## IV. Why Current Practice Cannot See These Failures

If the channel is where collaboration fails, then the field's measurement and training practice is aimed at the wrong target. Most existing benchmarks evaluate the model in isolation, and every major release reports solo performance as the primary evidence of capability [30], [31]. Recent work has begun to challenge this paradigm, showing that models superior in isolation may offer little benefit when a human is involved [32], that human-AI teams often underperform either party alone [33], and that complementary team performance is rarely achieved [34].

Even these studies measure outcome accuracy on short, self-contained tasks, whereas a developer architecting a system over dozens of turns faces a different challenge than a user answering a math question. Such factors, whether the human kept up, whether they noticed when the AI drifted, and whether they could later reconstruct how a decision was made, are rarely captured by mainstream model evaluations. A benchmark reporting only final accuracy scores a session in which drift was caught immediately the same as one where it was never caught but the answer happened to be right. The failures documented in Section III are thus largely invisible to current instruments.

Training practice has the same blind spot, and for a similar reason. Many widely used alignment methods primarily reward properties of outputs, with limited direct supervision of the collaboration process [35]. Such objectives can incentivize outputs that appear correct without guaranteeing the correctness or transparency of the intermediate reasoning, including by understating uncertainty. Because reward attaches to the final output, wrong intermediate steps go unpenalized, making it harder for collaborators to trace reasoning, spot mistakes, or intervene [27], [36]. Without incentives for transparency, there is no guarantee that models will reliably exhibit it during collaboration. We argue that this is at least partly a channel-level property rather than solely a model-level one, and that it is unlikely to be resolved by scale alone.

## V. Alternative Views

We address several objections to the position taken in this paper, each of which we take to be reasonable rather than easily dismissed.

### *A. Collaboration failures stem from interface design, user expertise, or workflow design, not the channel*

*Response:* These factors matter, and we do not claim the channel is the sole cause. However, each operates through the channel: interfaces shape pacing, expertise sets the human's comprehension rate, and workflows determine where verification occurs. Treating the channel as a research object unifies these factors instead of competing with them.

### *B. Sycophancy and drift will be solved by bigger and better models, making synchronization unnecessary [37]*

*Response:* Better models reduce sycophancy in controlled settings, but they also produce longer, denser, and more confident outputs [38], making the human's job harder, not easier [39]. A model that drifts less but produces responses the human cannot fully process does not solve the synchronization problem; it relocates it.

### *C. Adding synchronization friction reduces productivity and slows adoption; adaptive autonomy outperforms fixed checkpoints [40], [41]*

*Response:* Generating code or text faster is not the same as being more productive [42]; the quality and accountability behind the output also matter. A system producing 1,000 lines of unverified code is not clearly more productive than one producing 200 lines the human understands [43], [44]. The relevant design question is not whether to add friction but where to place it: simple tasks should flow freely, while high-stakes decisions warrant verification.

### *D. Existing tools such as Claude Code and Codex already provide accept/reject mechanisms for AI-generated edits [1], [2]*

*Response:* These tools offer accept/reject for edits, but users routinely enable automatic approval to avoid friction, and a gate that is always open is not a gate. Approval at the level of a dialog box asks the human to click, not to engage with the claim. We argue that effective synchronization should extend into the content itself, rather than relying solely on interface-level approval; such content-level mechanisms are not yet standard in current interfaces.

### *E. This is a human-computer interaction problem, not an ML research problem; the ML community should focus on models and HCI should design interfaces*

*Response:* Synchronization is a protocol property, not a presentation layer. Just as a server cannot simply "wrap" TCP, a model has to participate natively: producing output at a granularity the human can check, exposing faithful dependencies between claims, and signaling uncertainty. That is a training objective, not a parser, and the channel is bidirectional, since human feedback should condition future generation. Producing claims a human can verify, in a structure a human can follow, may need to be supported at the model and training level rather than relying exclusively on post-hoc interface mechanisms. This is precisely why the channel itself must become a first-class research object.

## VI. Conclusion

AI is taking on consequential roles, from writing code and operating vehicles to advising legal cases and making medical recommendations. A physician who cannot explain an AI treatment recommendation, or a developer who cannot trace how an AI-generated system was designed, reflects a channel that lacks synchronization, and errors from unverified decisions compound at scale.

These failures can arise not only from model deficiencies, but also from inadequate pacing, error recovery, and shared-state mechanisms in the channel through which the two parties work together.

**Call to Action**

We invite the community to (1) develop benchmarks that evaluate the human-AI pair rather than the AI alone, measuring whether the human kept up, detected drift, and can reconstruct the decision history; (2) incorporate comprehension, drift detection, and traceability into training objectives alongside task accuracy; and (3) build and evaluate synchronization protocols giving humans structured control over the pace and direction of collaboration. Without research effort on the channel itself, we risk systems that are increasingly capable but decreasingly accountable. We mark the shift to a new standard: *human-understanding-in-the-loop*.

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